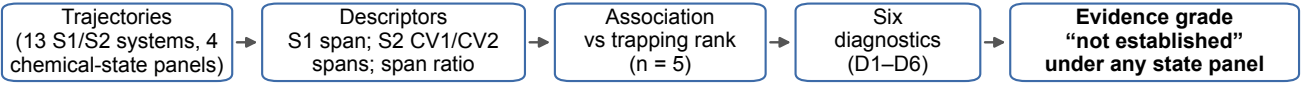

Six diagnostics → outcomes in this study (n = 5, tie-aware exact permutation)

D1 Windowed estimator variance	noise vs separations (2.6–5.3 kcal/mol); unresolvable at 26 ns
D2 Estimator–consistency matrix	no cell significant (all $p \geq 0.267$); sign unstable across estimators
D3 Convergence–timescale scaling	no decay over 2–24 ns windows; no extrapolated sampling length
D4 Ordinal sensitivity model	no decision-level cell ($P \leq 0.78$ across assumed bands)
D5 Counterfactual sensitivity	leave-one-out −0.63…+0.63; single-point exclusion flips sign
D6 Chemical-state audit	corrections move ρ −0.67 → −0.05; state vs noise unresolved

Verdict: no robust association between the computed spans and trapping , not established under any chemically defensible ligand-state panel. The output of this work is the audit workflow itself.